Performance Testing – Comic Craft

AI Comic Story Creator using Gemini Models

Field	Details
Date	01 October 2026
Team id	To be filled by team
Project name	Comic craft-AI comic story creator using gemini models
Maximum marks	5 Marks
Testing tools used	Postman/fast API interactive API documentation(/docs)
Type of testing	Performance testing of comic-generation and image -generation API workflows
Target module /API	/generate,/generate-comic /json,/test-image
Test environment	Local Fast API development server
Test date	To be filled with actual execution date

Step 1: Testing Overview

Performance testing is used to evaluate how Comic Craft behaves when its comic-generation workflow is invoked. The project uses Fast API for routing, Gemini Flash for a structured five-panel outline, Gemini Pro for narration and dialogue, Stable Diffusion for panel illustrations, and FPDF for PDF export. The documented workflow is therefore tested at the API and generation-module level.

Step 2: Test Scenarios

s.no	Test scenario/description	No .of virtual users	Duration(sec)	Expected outcomes
1	Generate a comic through /generate using story prompt, character	Testing to me and art style	To be recorded	Comic preview is generated successfully
2	Generate a comic through /generate -comic/json using and prompt	To test reply load.	To be recorded	Json response contains comic layout data
3	Invoke/test-image with a custom image prompt.	To be recorded	To be recorded	Stable diffusion generates and saves there
4	Repeat comic generation workflow to observe behaviour under repeat	To be request	To be recorded	Requests complete without unexpected fail

Step 3: Performance Test Results

s.no	Metric	Target value	Actual value	Status(pass/fail)	Remarks
1	Response time(avg)	<2 seconds	Not measured in source	Not evaluated	Requirement execution of a

					performance test
2	Response time(max)	<5 second	Not measured in source	Not evaluated	Requires execution of a performance test
3	Throughput(req/sec)	Project - specific target	Not measured in source	Not evaluated	Requires load of a test execution
4	Error rate	<1%	Not measured in source	Not evaluated	Requires observed request /error counts.
5	Cpu utilization	<80%	Not measured in source	Not evaluated	Requires server monitoring during test.
6	Memory utilization	<80%	Not measured in source	Not evaluated	Requires server monitoring during test.

Step 4: Observations & Analysis

1. Comic Craft exposes browser-based and JSON-based generation flows through Fast API routes.
2. The /generate workflow chains outline generation, story generation, image generation, layout building and PDF export; therefore, total response time depends on the complete AI-generation pipeline.
3. The /generate-comic/json route follows a similar generation pipeline and returns layout data with the generated PDF path
4. The /test-image route provides an isolated way to test image-generation performance.
5. The project documentation specifies performance-test target metrics such as average response time, maximum response time, throughput, error rate, CPU utilization and memory utilization. Actual measurements are not present in the supplied Performance Testing template.

Step 5: Screenshots / Evidence

Attach actual screenshots of the selected performance-testing tool here. Recommended evidence includes: request configuration, concurrent-user/load configuration, response-time results, throughput, error summary, and CPU/memory monitoring during Comic Craft execution. The supplied source template explicitly requests screenshots or tool output as evidence

Project Workflow Reference

Comic Craft accepts a story prompt, character name, setting, story tone and art style; generates a structured five-panel outline, expands it into narration/dialogue, generates illustrations, builds

the panel layout, and exports the completed comic as a PDF. The project runs locally with Uvicorn/Fast API and exposes interactive API documentation at /docs.